

Evidence-Driven Dataset Intelligence and Recommendation System

LLM-assisted discovery, content-aware suitability estimation, long-tail exploration, provenance, calibrated confidence, and explainable ranking

Research-Grounded Proposal — Version 2 (Review-Integrated Revision)

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Status of this document. This is Version 2 of the Capstone Project 1 research-grounded proposal. It incorporates a structured technical review of the Version 1 proposal and closes nine identified gaps: (1) a formalized evidence-resolution algorithm, (2) quantitative bias and calibration metrics, (3) an LLM zero-shot baseline, (4) a cold-start transfer-estimation fallback, (5) a human evaluation protocol for trust and explanation quality, (6) an early compute-budget validation milestone, (7) explicit team ownership across five to six engineers, (8) a legal/ToS compliance checklist for automated sample acquisition, and (9) a minimum-viable CLI moved into core scope. Sections carrying **[NEW]** or **[REVISED]** are new or materially revised relative to Version 1; unmarked sections are retained with light editorial changes for consistency.

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1 Executive Summary

The project is an evidence-driven dataset intelligence system for Machine Learning (ML), Deep Learning (DL), and Computer Vision (CV) practitioners. A user supplies a natural-language description of a project. The system converts the description into a structured task specification, discovers candidate datasets across multiple public repositories, gathers evidence about those datasets, performs lightweight analysis of accessible samples where feasible, estimates task-specific utility, mitigates popularity bias, and produces a diversified, explainable recommendation set.

Version 1 of this proposal established that a metadata-plus-retrieval recommender is, by itself, insufficiently differentiated: dataset recommendation is already a studied information-retrieval problem, and a trained bi-encoder retriever (DataFinder) already demonstrates strong gains over keyword search [1]. Version 1 therefore repositioned the project around *evidence-backed selection* rather than retrieval alone, introducing an evidence ledger, content-aware fingerprinting, task-utility probing, and popularity-bias mitigation.

This revision (Version 2) closes the remaining gap between that design intent and a defensible, gradable, and potentially publishable research artifact. The single highest-leverage addition is a **formal evidence-resolution algorithm** (Section 8.2) that derives the system’s trust labels (*Verified*, *Corroborated*, *Single-source claim*, *Conflicting*, *Unknown*) from a Dempster–Shafer belief-combination rule over source reliabilities, rather than assigning them heuristically. This is the part of the system that is not a re-application of DataFinder, DataPerf, Cleanlab, Croissant, or GIST — it is new, mathematically specified, and independently evaluable. Around this core, Version 2 adds quantitative bias and calibration metrics (Section 12.1), a cold-start fallback for datasets that cannot be sampled (Section 11.1), a human trust-evaluation protocol (Section 18.4), an LLM zero-shot baseline (Section 19), and practical project-management additions (Sections 20.2 and 20.3) needed to execute the plan with a five-to-six-person team on Apple Silicon hardware.

Central thesis (unchanged). A dataset recommender should answer not only “What datasets look relevant?” but “Which datasets are most defensibly useful for this task, what evidence supports that judgment, how uncertain is it, and what useful alternatives would otherwise be missed?”

2 Project Foundation Retained from Version 1

The system continues to target ML/DL/CV users, accepts natural-language project descriptions, operates across multiple public dataset repositories (Kaggle, Hugging Face Hub, UCI, OpenML, and others), avoids requiring complete dataset downloads in the default path, and returns a ranked, explained set of recommendations rather than an unranked search-result list.

3 Research Gap and Problem Definition

3.1 Why a metadata-only recommender is insufficient

Dataset recommendation is already established as an information-retrieval problem. DataFinder constructs a benchmark from research needs and evaluates BM25, nearest-neighbor, and dense bi-encoder retrieval, reporting substantial gains from a trained bi-encoder over generic search approaches [1]. A project whose main novelty is “LLM extraction + vector retrieval + metadata scoring” would risk reproducing an existing direction. The system’s role is therefore shifted from pure retrieval toward evidence-backed selection: metadata supports fast candidate generation and

constraint filtering, while additional evidence is drawn from dataset structure, documentation, provenance, and small accessible samples before stronger suitability claims are made.

3.2 Why repository metadata cannot be treated as ground truth

Repository metadata is publisher-supplied. Documentation quality is uneven, and dataset visibility can itself correlate with documentation quality. Licensing is especially sensitive: public reporting on dataset licensing quality documents substantial rates of missing or incorrectly stated licenses across widely used public ML datasets [7]. The system therefore distinguishes a repository *claim* from a verified or cross-supported *fact*. “License: CC BY 4.0” may be reported as a repository claim; it must not be silently translated into “commercially safe.” [REVISED] Version 2 makes this distinction computable rather than descriptive — see Section 8.2.

3.3 Why popularity must not define suitability

Downloads, stars, ratings, and citations are useful community evidence but are not equivalent to task fitness. A popularity-driven ranking favors datasets with stronger ecosystem visibility and richer metadata, creating a feedback loop in which well-known datasets become increasingly discoverable while specialized or smaller datasets remain underexposed. The final ranking treats popularity as contextual evidence, not core quality; a long-tail exploration mechanism searches for relevant candidates that are not dominant on popularity signals, and the final set is diversified so that several near-identical popular datasets do not crowd out useful alternatives. [REVISED] Version 2 requires this claim to be measured, not merely designed for — see Section 12.1.

4 Research Questions and Objectives

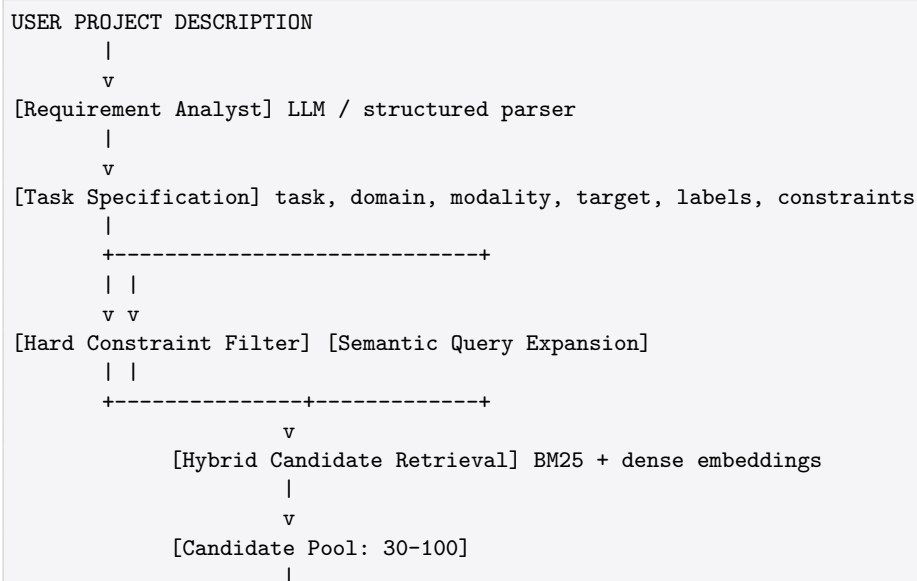
1. Build a normalized multi-source dataset corpus with stable identifiers and provenance pointers.
2. Extract structured task requirements using an LLM or a deterministic parser constrained by a schema.
3. Generate candidates through hybrid retrieval and hard-constraint filtering.
4. Assemble an evidence ledger for metadata, documentation, provenance, license claims, repository signals, and sample findings, resolved by a **formal combination rule** ([NEW] Section 8.2) rather than by heuristic labeling.
5. Compute content-aware dataset fingerprints from small accessible samples when technically and legally feasible, with an explicit **transfer-estimate fallback** ([NEW] Section 11.1) when sampling is infeasible.
6. Estimate task-specific utility using lightweight probes rather than full model training.
7. Rank candidates with an evidence-aware, uncertainty-aware scoring model and diversify the final set, reporting the resulting **exposure disparity** ([NEW] Section 12.1) relative to a non-diversified baseline.
8. Produce explanations that separate claims, measured evidence, assumptions, unknowns, and limitations, and validate that these explanations measurably change user trust ([NEW] Section 18.4).

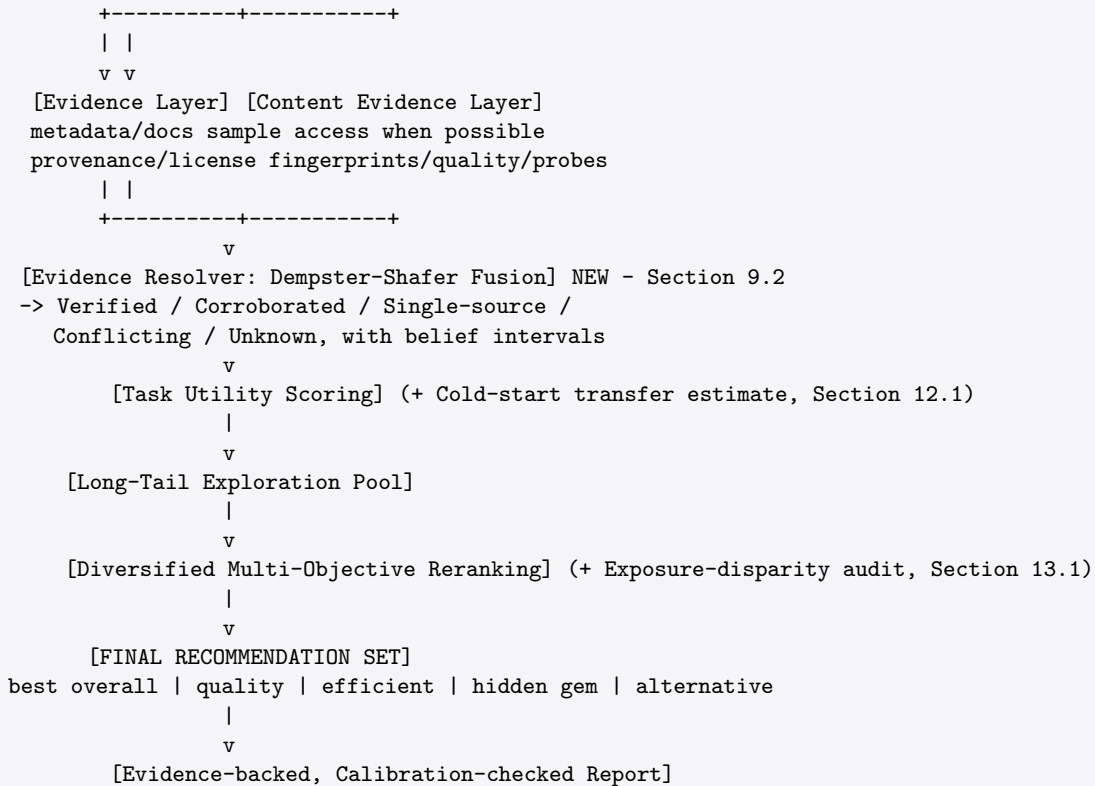
9. Evaluate the system against strong retrieval baselines — including a zero-shot LLM baseline ([NEW] Section 19) — robustness tests, calibration analysis, and a purpose-built long-tail benchmark.

5 Final Design Principles

- **Evidence over assertion:** a recommendation factor must be traceable to a source, observation, computed metric, or explicit assumption.
- **Suitability over popularity:** popularity can inform confidence or community evidence but should not dominate task-specific utility.
- **Long-tail discovery is intentional:** the system must preserve a candidate path for relevant datasets with weak visibility.
- **Content when available:** actual sample evidence supplements repository descriptions instead of requiring complete downloads.
- **Hard constraints are filters:** mandatory requirements (modality, minimum size, usage constraints) are not merely score penalties.
- **Unknown is a valid state:** missing evidence lowers confidence rather than silently becoming a neutral or favorable value.
- **Recommendations are sets, not only rankings:** the top five should be useful together and avoid redundant dataset families.
- **Local-first and reproducible:** the core pipeline runs with open-source components and modest local compute, with optional acceleration where available.
- **[NEW] Confidence must be calibrated, not decorative:** an “Evidence strength: High” label is only meaningful if it is empirically shown to correlate with correctness. See Section 12.1.

6 Final System Architecture





6.1 Candidate-pool strategy

The system retrieves a broader candidate pool than the final recommendation set. A recommended default is 50–100 candidates after retrieval and hard filtering, followed by deeper evidence analysis on a smaller subset. This keeps the expensive content-aware stages bounded, and its feasibility is validated explicitly in Milestone M0 (Section 20.2) before the team commits engineering effort around this number.

7 Canonical Dataset Representation

The dataset corpus uses a canonical internal record so that Kaggle, Hugging Face, UCI, OpenML, and other repositories can be compared without pretending that their native schemas are identical. Croissant is used where available as a standards-aligned metadata representation, while repository-specific fields remain attached to their source rather than being flattened away.

8 Evidence Ledger and Trust Model

The evidence ledger is a core project component. Each important dataset claim is stored with its source, observation time, evidence type, confidence, and any contradiction, preventing the explanation layer from turning uncertain metadata into authoritative-sounding statements.

License principle. The system does not present automated license parsing as legal clearance. It classifies, compares, flags conflicts, and exposes the evidence path; legal interpretation remains outside the model’s authority.

8.1 Evidence states (Version 1 design)

Version 1 specified five qualitative states: *Verified*, *Corroborated*, *Single-source claim*, *Conflicting*, and *Unknown*. These states were not previously derived from a stated combination rule.

8.2 [NEW] Formal Evidence Resolution via Dempster–Shafer Fusion

This is the primary technical contribution of the project and the component most distinct from prior work (DataFinder, DataPerf, Cleanlab, Croissant, GIST each address a different sub-problem; none formalizes multi-source claim resolution for dataset trust). We adopt Dempster–Shafer evidence theory [10, 11] because, unlike a simple weighted average, it natively represents *ignorance* as a first-class quantity distinct from disagreement — which matches the system’s requirement that “Unknown” and “Conflicting” be different states rather than both collapsing to a mid-range score.

8.2.1 Frame of discernment and basic probability assignment

For a single claim c about a dataset d (for example, “license = CC BY 4.0”, or “modality = image”), define the binary frame of discernment $\Theta = \{\top, \perp\}$ (claim true / claim false). Let $\mathcal{S}(c)$ be the set of evidence sources that address c (a source that is silent on c contributes nothing and is excluded). Each source $i \in \mathcal{S}(c)$ has a prior reliability $r_i \in (0, 1)$ (Table 1) and a stance $\sigma_i \in \{\text{assert-true}, \text{assert-false}\}$. Its basic probability assignment (BPA) over 2^Θ is:

$$m_i(\top) = \begin{cases} r_i & \sigma_i = \text{assert-true} \\ 0 & \sigma_i = \text{assert-false} \end{cases} \quad m_i(\perp) = \begin{cases} 0 & \sigma_i = \text{assert-true} \\ r_i & \sigma_i = \text{assert-false} \end{cases} \quad (1)$$

$$m_i(\Theta) = 1 - r_i \quad (2)$$

The mass placed on Θ represents the source’s acknowledged unreliability: a source is never treated as fully authoritative, so $r_i < 1$ strictly.

Source type	Example claim it can support	Prior
Directly measured from sample	modality, approximate class count, format	0.9
Peer-reviewed paper describing the dataset	task, domain, collection method	0.8
Croissant-validated metadata record	license, schema, splits	0.8
Official dataset card (repository-hosted)	license, size, task, modality	0.7
Bare repository metadata field	license, tags, size	0.6
Unverified community description (forum, README of unclear origin)	any	0.3

Table 1: Default source-reliability priors. These are hand-set initial values for the prototype and are tuned post hoc using the calibration procedure in Section 12.1.

8.2.2 Combination rule

Sources are combined pairwise using Dempster’s rule of combination. For two BPAs m_1, m_2 over Θ :

$$m_{12}(A) = \frac{1}{1 - K} \sum_{\substack{B, C \subseteq \Theta \\ B \cap C = A}} m_1(B) m_2(C), \quad A \neq \emptyset \quad (3)$$

$$K = \sum_{\substack{B, C \subseteq \Theta \\ B \cap C = \emptyset}} m_1(B) m_2(C) \quad (4)$$

where K is the *conflict mass* produced by combining m_1 and m_2 . Sources in $\mathcal{S}(c)$ are combined iteratively; Dempster’s rule is commutative and associative, so the combination order does not affect the final result, though numerical care (accumulating $\log(1 - K)$ terms) is used to avoid underflow when $|\mathcal{S}(c)|$ is large. After combining all sources, the system retains:

- **Belief:** $\text{Bel}(\top) = m(\top)$ — the mass of evidence directly supporting the claim.
- **Plausibility:** $\text{Pl}(\top) = m(\top) + m(\Theta) = 1 - m(\perp)$ — the upper bound consistent with the evidence.
- **Uncertainty interval:** $[\text{Bel}(\top), \text{Pl}(\top)]$.
- **Cumulative conflict:** K_{total} , the total conflict mass accumulated across all pairwise combinations, capturing how much the sources actively disagreed (as opposed to one source simply being silent).

8.2.3 Mapping to trust states

The qualitative labels used throughout the system’s explanations are now *derived* quantities rather than assigned labels:

$$\text{Verified} \iff \text{Bel}(\top) \geq \tau_v \wedge K_{\text{total}} \leq \kappa_{\text{low}} \wedge |\mathcal{S}(c)| \geq 2 \quad (5)$$

$$\text{Corroborated} \iff \tau_c \leq \text{Bel}(\top) < \tau_v \wedge K_{\text{total}} \leq \kappa_{\text{low}} \wedge |\mathcal{S}(c)| \geq 2 \quad (6)$$

$$\text{Single-source claim} \iff |\mathcal{S}(c)| = 1 \quad (7)$$

$$\text{Conflicting} \iff K_{\text{total}} > \kappa_{\text{high}} \quad (8)$$

$$\text{Unknown} \iff |\mathcal{S}(c)| = 0 \quad (9)$$

with default thresholds $\tau_v = 0.85$, $\tau_c = 0.60$, $\kappa_{\text{low}} = 0.10$, $\kappa_{\text{high}} = 0.30$ (tunable; see calibration in Section 12.1).

A fully worked numeric example (three sources disagreeing on a license claim) is given in Appendix D. This algorithm feeds directly into the Evidence term $\gamma \cdot \text{Evidence}$ of the ranking model in Section 14: **Evidence**(d) is defined as the mean $\text{Bel}(\top)$ across all claims resolved for d , penalized by mean K_{total} , giving a single computed number rather than a qualitative impression.

9 Content-Aware Dataset Fingerprinting

When a repository permits lightweight sample access, the system creates a Dataset Fingerprint: not a single vector claiming to summarize the entire dataset, but a compact evidence package describing what was actually observed in the inspected sample. Sample size is adaptive rather than fixed — a small initial sample establishes feasibility, and the system expands the sample only when a signal is unstable and the additional computation is justified.

10 Data Quality Analysis

Cleanlab-style model-assisted checks are included as a tool, not as the definition of correctness. Cleanlab’s confident-learning workflow requires suitable out-of-sample model probabilities and can identify examples with suspicious labels [3]. These outputs are aggregated into evidence features such as estimated label-issue rate or the share of high-risk examples.

Algorithm 1 Evidence Resolution for a Single Claim

```

1: function RESOLVECLAIM( $c$ , sources  $\mathcal{S}(c)$ )
2:   if  $|\mathcal{S}(c)| = 0$  then
3:     return (UNKNOWN, Bel = 0, Pl = 1)
4:   end if
5:    $m \leftarrow$  BPA of the first source in  $\mathcal{S}(c)$ 
6:    $K_{\text{total}} \leftarrow 0$ 
7:   for each remaining source  $i \in \mathcal{S}(c)$  do
8:      $m_i \leftarrow$  BPA( $i$ ) ▷ using  $r_i$  from Table 1
9:      $K \leftarrow$  conflict mass of combining  $m$  with  $m_i$ 
10:     $m \leftarrow$  Dempster-combine( $m$ ,  $m_i$ )
11:     $K_{\text{total}} \leftarrow K_{\text{total}} + K \cdot (1 - K_{\text{total}})$  ▷ running conflict, bounded in  $[0, 1]$ 
12:   end for
13:   Bel  $\leftarrow m(\top)$ ; Pl  $\leftarrow 1 - m(\perp)$ 
14:   if  $|\mathcal{S}(c)| = 1$  then
15:     return (SINGLESOURCECLAIM, Bel, Pl)
16:   else if  $K_{\text{total}} > \kappa_{\text{high}}$  then
17:     return (CONFLICTING, Bel, Pl)
18:   else if Bel  $\geq \tau_v$  and  $K_{\text{total}} \leq \kappa_{\text{low}}$  then
19:     return (VERIFIED, Bel, Pl)
20:   else if Bel  $\geq \tau_c$  and  $K_{\text{total}} \leq \kappa_{\text{low}}$  then
21:     return (CORROBORATED, Bel, Pl)
22:   else
23:     return (SINGLESOURCECLAIM, Bel, Pl) ▷ low-confidence fallback label
24:   end if
25: end function

```

1. Obtain a task-appropriate sample and preserve the dataset’s original labels.
2. Create out-of-sample prediction probabilities using cross-validation or a held-out procedure.
3. Run label-issue detection and record suspicious-example counts rather than silently deleting data.
4. Compute duplicate and near-duplicate statistics using hashes and embedding similarity.
5. Compute modality-specific structural diagnostics.
6. Store all measurements in the evidence ledger with sample size and procedure, feeding the DIRECTLYMEASURED source type in Table 1.
7. Use the resulting features as quality evidence with uncertainty bounds, not as ground truth.

11 Task-Specific Utility Estimation

The key extension beyond conventional dataset recommendation is a lightweight estimate of whether a dataset appears useful for the stated ML task. DataPerf demonstrates the value of evaluating data selection through downstream task performance [2]; the system adapts that principle under local-compute constraints. A probe score is an estimate under a chosen probe model, sample, and

evaluation split — it is not a guarantee that the full dataset will produce the same downstream performance.

11.1 [NEW] Cold-Start Fallback via Transfer Estimation

Section 17.5 (Sample Acquisition Procedure) already triages several sampling strategies, but Version 1 did not specify a behavior when *none* succeed — a common outcome for paywalled, gated, or terms-restricted repositories. Rather than silently dropping such datasets from utility scoring (which would bias the system back toward easily-scraped, and therefore already-popular, datasets), Version 2 defines a transfer estimate.

For an unsampled candidate d , let $\mathcal{N}_k(d)$ be its k nearest already-probed datasets in metadata/description embedding space (using the same encoder as retrieval). Define an RBF similarity weight $w(d, d') = \exp(-\|e(d) - e(d')\|^2 / (2\sigma^2))$ and estimate:

$$\hat{u}(d) = \frac{\sum_{d' \in \mathcal{N}_k(d)} w(d, d') \cdot u(d')}{\sum_{d' \in \mathcal{N}_k(d)} w(d, d')} \quad (10)$$

with an inflated uncertainty band that widens as the nearest probed neighbor becomes less similar:

$$\sigma_{\hat{u}}(d) = \sigma_{\text{base}} + \lambda \left(1 - \max_{d' \in \mathcal{N}_k(d)} w(d, d') \right) \quad (11)$$

Transfer estimates are explicitly tagged TRANSFERREDESTIMATE—NOTDIRECTLYOBSERVED in the evidence ledger and the explanation card, are excluded from ever producing a VERIFIED content label, and are down-weighted (not discarded) in the final ranking. This keeps restricted-but-relevant datasets in contention without overstating what was actually measured.

12 Popularity Bias and Long-Tail Discovery

Popularity is separated from suitability. The retrieval stage reserves candidate slots for datasets that meet task requirements but are not dominant on repository popularity signals; this exploration pool is then evaluated by the same evidence and utility criteria as mainstream candidates. The objective is not to reward obscurity — a lesser-known dataset reaches the final recommendation only when its evidence supports utility, diversity, efficiency, specialization, or another user-relevant advantage.

```
candidate_score = task_fit + evidence_quality + content_utility + feasibility
long_tail_pool = relevant_candidates
                    filtered to non-dominant popularity percentiles
                    while preserving hard constraints
final_set = diversified_selection(main_pool + long_tail_pool)
```

12.1 [NEW] Quantifying Bias: Exposure Disparity and Reliability Calibration

Version 1 asserted that diversification would reduce popularity bias and targeted “diversity metric improves by more than 20%.” A diversity increase alone does not prove that popularity bias was reduced — it proves that the top- k set spread out. Version 2 adds two measurements that are actually tied to the stated research claims.

12.1.1 Exposure Disparity Index (EDI)

Using the benchmark of Section 18.1, bucket the full retrieved candidate pool for each query into popularity deciles $j \in \{1, \dots, 10\}$ (by downloads/stars, log-scaled). Let R_j be the fraction of *human-judged-relevant* candidates falling in decile j , and let E_j be the fraction of *final recommendation slots* (aggregated over the benchmark) drawn from decile j . Define:

$$\text{EDI} = \frac{1}{2} \sum_{j=1}^{10} |E_j - R_j| \quad (12)$$

This is the total variation distance between where recommendations actually land and where relevance says they should land. $\text{EDI} = 0$ means exposure exactly tracks relevance, regardless of popularity; $\text{EDI} \rightarrow 1$ means exposure is maximally distorted by popularity. The system reports EDI for (a) bi-encoder-only ranking, and (b) the full system with long-tail exploration and diversified reranking, and the hypothesis is $\text{EDI}_{\text{full}} < \text{EDI}_{\text{baseline}}$. A secondary sanity-check statistic, the Gini coefficient of the popularity-decile distribution of recommended items, is reported alongside EDI but is not the primary claim, since a lower Gini coefficient alone does not distinguish “spread evenly across relevant items” from “spread evenly regardless of relevance.”

12.1.2 Calibration of confidence labels

The design principle “confidence must be calibrated, not decorative” is tested using standard calibration diagnostics against the benchmark’s human relevance/quality judgments. Predicted confidence $\text{Bel}(\top)$ values (Section 8.2) are grouped into B equal-width bins; within each bin b , $\text{acc}(b)$ is the observed fraction of claims confirmed correct by the benchmark’s ground truth, and $\text{conf}(b)$ is the mean predicted belief in that bin. Expected Calibration Error:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)| \quad (13)$$

and, for continuous utility scores $p_i \in [0, 1]$ against binary outcomes o_i (dataset judged suitable or not by a probe/human), the Brier score:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (p_i - o_i)^2 \quad (14)$$

Both are reported as reliability diagrams (predicted confidence vs. observed accuracy) in the final report, following standard practice for calibration evaluation of predictive models [12]. Low ECE and low Brier score are required before “Evidence strength: High” is allowed to appear in any user-facing explanation card; if calibration is poor, the source-reliability priors in Table 1 are the first tuning target.

13 Diversified Recommendation with Utility and Diversity

GIST formulates a max-min diversification problem that balances submodular utility with diversity and provides a $1/2$ -approximation guarantee for the stated optimization problem [5]. The project adopts the principle rather than requiring full-scale implementation over millions of items. After candidate analysis, the system operates on a bounded pool and uses a GIST-inspired greedy procedure or a computationally lighter farthest-first/submodular alternative. Dataset-family similarity

is considered in diversification: mirrors, versions, and closely derived datasets should not occupy multiple final slots when they offer effectively the same evidence and coverage.

14 Final Multi-Objective Recommendation Model

The final recommendation model is multi-objective rather than a single metadata-weighted average:

$$R(D \mid Q) = \alpha \cdot \text{Fit} + \beta \cdot \text{Utility} + \gamma \cdot \text{Evidence} + \delta \cdot \text{Coverage} + \varepsilon \cdot \text{Novelty} - \lambda \cdot \text{Risk} \quad (15)$$

where **Fit** captures explicit requirements, **Utility** captures content-aware task evidence (with transfer estimates from Section 11.1 where direct probing is infeasible), **Evidence** is now the computed quantity $\text{Bel}(\top)$ averaged across claims and penalized by K_{total} (Section 8.2), **Coverage** captures diversity and representativeness, **Novelty** rewards useful long-tail discovery, and **Risk** captures uncertainty, contradiction, missing governance information, or operational problems.

The coefficients $(\alpha, \beta, \gamma, \delta, \varepsilon, \lambda)$ are not assumed universally correct. Initial values are hand-designed for the prototype and subsequently tuned on the benchmark (Section 18.1) using pairwise human relevance judgments and a simple learning-to-rank model (logistic regression over the six components, or gradient-boosted pairwise ranking if enough judgment pairs are collected). Popularity is intentionally excluded from the core suitability term; it may enter only as contextual evidence, long-tail stratification, or a controlled bias-audit variable (as used directly in Section 12.1).

15 Explanation and Recommendation Output

Each recommended dataset produces an evidence-backed recommendation card rather than a single score:

```
Dataset: <name>
Overall recommendation: 91/100 (illustrative structure)
Best for: <task/domain/modality>
Evidence strength: High / Medium / Low (derived from Bel(true), Section 9.2 - calibrated, Section 13.1)

Why it fits
- Task: measured / corroborated / source claim
- Domain: ...
- Modality: ...
- Labels: ...
- Observed sample quality: ...

Why it is not perfect
- Unknown or conflicting evidence: ...
- Operational limitations: ...
- [If applicable] Utility is a transferred estimate, not directly observed (Section 12.1)

Why it appears here
- More diverse than the top popular alternatives
- Smaller but cleaner / better balanced

Sources and timestamps
- Repository metadata
- Dataset documentation / paper
- Sample analysis
```

This card is the artifact evaluated directly in the human trust study (Section 18.4).

16 Local Implementation Strategy

The core implementation targets ordinary personal computers, specifically the team’s available hardware (MacBook Air M1, 8GB RAM). The system is CPU-first, memory-conscious, cache-heavy, and modular. Optional acceleration can be enabled when a supported local device provides it. The project does not depend on proprietary APIs for its central pipeline.

16.1 Environment setup procedure

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip

pip install pandas numpy scipy scikit-learn
pip install sentence-transformers
pip install faiss-cpu
pip install cleanlab
pip install mlcroissant
# Optional retrieval / diversification components
pip install pyserini
pip install submodlib
```

Package versions are pinned after the first successful environment build. The repository contains a requirements/lock file and a small environment-validation script.

16.2 Repository ingestion procedure

1. Create one source adapter per repository; never let repository-specific JSON fields leak directly into downstream ranking logic.
2. Assign a stable internal dataset identifier and retain the native source identifier.
3. Normalize task, modality, format, class count, sample count, license claim, documentation links, timestamps, and popularity signals into the canonical schema.
4. Retain raw source snapshots or request/response files for reproducibility where licensing and storage terms permit (see Section 16.3 below).
5. Attempt Croissant parsing when a valid record is available; retain provenance links and validation results.
6. Generate the retrieval document from a controlled concatenation of name, description, tags, task semantics, and structured textual fields.
7. Precompute dense embeddings and store them in the local vector index.
8. Build a BM25 index over the same normalized textual representation.
9. Cache every source fetch so experiments do not repeatedly hit external services.

16.3 [NEW] Legal and ToS Compliance Checklist for Automated Sample Acquisition

Section 17.5’s sample-acquisition ordering (streaming/range requests, sample endpoints, previews, manual acquisition) is a technical strategy; Version 1 did not separately confirm that each strategy is permitted by the relevant platform. Before any automated fetch of a repository, the ingestion adapter for that source must record:

1. Whether the platform’s API terms of service permit automated/programmatic access for the intended use (research/educational, non-redistributive).
2. The platform’s documented rate limits, and a corresponding client-side rate limiter (never inferred empirically by trial and error against the live service).
3. Whether `robots.txt` or an equivalent access policy restricts the specific endpoints used.
4. Whether the dataset’s own license permits the sampling/inspection the system performs, independent of the platform’s API terms (a dataset can be API-accessible but license-restricted).

This checklist is stored per source adapter (not per dataset) and is reviewed once per repository integration, not once per request. The caching-first policy already specified in Version 1 additionally serves this compliance purpose: every source fetch is cached so experiments do not repeatedly re-request the same resource.

16.4 Requirement extraction procedure

```
{
  "task": "image classification",
  "domain": "plant disease",
  "modality": "image",
  "learning_type": "supervised",
  "label_type": "multiclass",
  "min_samples": 5000,
  "required_classes": null,
  "license_requirement": "academic_use",
  "hard_constraints": [],
  "soft_preferences": []
}
```

The schema rejects unknown fields or places them into an explicit “unresolved constraints” field. This makes requirement-extraction errors measurable rather than invisible.

16.5 Hybrid retrieval procedure

1. Run BM25 for exact terms such as disease names, modalities, target variables, file formats, or domain-specific vocabulary.
2. Run dense retrieval on the full natural-language and structured query representation.
3. Fuse the candidate lists using reciprocal-rank fusion or calibrated score normalization.
4. Apply hard constraints after candidate generation so recall is not unnecessarily lost during the first lexical/dense pass.
5. Deduplicate mirrors and obvious dataset-family duplicates before deeper analysis.

6. Retain 30–100 candidates for the evidence stage and log the retrieval source that contributed each candidate.

16.6 Sample acquisition procedure

The system avoids indiscriminate full downloads. Preferred order: direct streaming or repository viewer APIs; small file/range requests where supported; public sample endpoints; repository-provided previews; finally, limited manual acquisition for benchmark datasets. Every sample-analysis record states the number of observations inspected and the acquisition method, and is checked against Section 16.3 before use.

16.7 Probe computation procedure

1. Feasibility check: can the candidate be sampled within a bounded size and time budget (validated empirically in Milestone M0, Section 20.2)?
2. If infeasible, invoke the transfer-estimate fallback (Section 11.1) instead of dropping the candidate.
3. Create a task-appropriate representation: image embeddings, text embeddings, or normalized tabular features.
4. Compute structural and distributional diagnostics.
5. If labels are available, train a small probe model with stratified validation and obtain out-of-sample predictions.
6. Run Cleanlab-style label-issue estimation where assumptions are satisfied.
7. Compute class coverage and representation diversity.
8. Persist all intermediate metrics so ranking can be rerun without recomputing samples.

16.8 Provenance and license workflow

```
Repository claim ----+
Dataset card -----+--> Evidence Resolver (Alg. 1) --> state: verified / corroborated /
Paper / source -----+ single-source / conflicting / unknown
Croissant -----+ with Bel(true), Pl(true), K_total
```

Automated license parsing remains a classification aid, not legal advice. Croissant 1.1 adds machine-actionable provenance concepts, vocabulary links, and usage-policy mechanisms [4]; the implementation adopts these structures where present and retains raw provenance references rather than flattening everything into a single “reliability score.”

17 End-to-End Recommendation Algorithm

```
def recommend(query, k=5):
    spec = extract_requirements(query)

    lexical = bm25_retrieve(spec.text_query, top_n=100)
    dense = dense_retrieve(spec.embedding_query, top_n=100)
    candidates = fuse_and_deduplicate(lexical, dense)
```



```

candidates = hard_filter(candidates, spec.hard_constraints)

raw_evidence = collect_evidence(candidates) # per-claim sources
evidence = {d: resolve_claims(raw_evidence[d]) for d in candidates} # NEW: Algorithm 1

feasible = select_sample_accessible(candidates, budget=probe_budget)
content = analyze_samples(feasible, spec)
unsampled = [d for d in candidates if d not in feasible]
transferred = transfer_estimate(unsampled, content) # NEW: Section 12.1

for d in candidates:
    d.fit = requirement_fit(d, spec)
    d.utility = content.get(d.id, transferred.get(d.id))
    d.evidence = mean_belief(evidence[d.id]) # from Algorithm 1
    d.risk = uncertainty_and_conflict(evidence[d.id])
    d.novelty = long_tail_value(d)
    d.coverage = coverage_value(d, spec)

pool = multi_objective_rank(candidates) # R(D/Q), Section 15
main, long_tail = split_exploration_pool(pool)
final = diversified_select(main + long_tail, k=k)
audit_exposure_disparity(pool, final) # NEW: Section 13.1, logged not blocking
return explain_with_evidence(final, evidence, content)

```

18 Evaluation Framework

Evaluation is treated as a first-class research artifact. The system is not considered successful merely because recommendations appear plausible; it must demonstrate where each additional component changes retrieval, ranking, discovery, or trustworthiness.

18.1 Hidden-Gem benchmark

A dedicated benchmark contains queries for which expert-approved datasets include at least one lesser-known candidate. The benchmark records relevance judgments and popularity strata before the final system is run. A hidden gem is not defined as “low downloads”; it is a dataset with useful relevance or task evidence whose visibility is lower than dominant alternatives. This benchmark is also the ground truth used for the EDI and calibration measurements in Section 12.1.

18.2 Robustness benchmark

Synthetic corruption tests remove or contradict selected metadata fields. For example, class counts are removed, licenses are marked unknown, and descriptions are weakened. The system should not collapse because one metadata source is incomplete; it should lower confidence, rely on other evidence where available, and expose uncertainty. With the formal resolver (Section 8.2), this becomes a directly testable property: removing a corroborating source should shift a claim from VERIFIED to CORROBORATED or SINGLE-SOURCE CLAIM in a predictable direction, which can be asserted as a unit test on Algorithm 1 rather than only observed qualitatively.

18.3 Avoiding fabricated result claims

No performance values are assumed in advance. Any target (e.g., a percentage gain) is a research hypothesis, not a result. Final tables are populated only after experiments are run. The project reports confidence intervals or repeated-run variability where appropriate.

18.4 [NEW] Human Evaluation of Trust and Explanation Quality

The system’s central thesis is a claim about *defensibility and trust*, which automated IR and diversity metrics cannot directly validate. A small human study is added as a required, not optional, evaluation component.

Design. Recruit 10–15 participants (peers/labmates with ML background). For a fixed set of 8–10 query–dataset pairs drawn from the benchmark, each participant sees, in randomized order: (a) a plain metadata card (name, description, raw license field, download count), and (b) the evidence-backed recommendation card from Section 8.2/12 for the same dataset. For each, participants rate on a 5-point Likert scale: (i) “I would trust this recommendation enough to start using the dataset,” and (ii) “I understand why this dataset was or was not a good fit.” Participants are also asked a forced-choice question: “which card would you act on faster?”

Analysis. Report mean Likert scores per card type with a paired comparison (e.g., Wilcoxon signed-rank test given the small, non-normal sample), and the forced-choice win rate. This directly tests the “evidence over assertion” design principle rather than asserting it.

19 Baselines and Ablation Plan

1. [NEW] **LLM zero-shot recommendation** — ask a general-purpose LLM directly for dataset recommendations with no retrieval, evidence, or probing. This is the practical alternative most users would otherwise reach for, and the project must show measurable gains over it (particularly on hidden-gem recall, license correctness, and hallucination rate — does the LLM recommend datasets that do not exist or misstate their license?).
2. BM25-only retrieval.
3. Dense embedding-only retrieval.
4. DataFinder-style trained bi-encoder retrieval.
5. Hybrid retrieval without evidence scoring.
6. Hybrid + metadata suitability scoring.
7. Hybrid + evidence ledger + Dempster–Shafer resolution, no uncertainty-aware ranking.
8. Hybrid + content-aware utility estimation (including transfer estimates).
9. Full system without diversification.
10. Full system with long-tail exploration and diversified reranking.

This sequence isolates whether each design layer contributes independently and prevents the final system from being presented as a monolithic black box.

20 Scope and Development Roadmap

20.1 Scope control

- The core MVP does not require training a large language model.
- The core MVP does not require complete downloads of every candidate dataset.
- Bi-encoder fine-tuning is optional until baseline retrieval is established.
- Full GIST reproduction at web-scale is not required; the project uses a bounded implementation or an efficient approximation.
- A single modality is used for initial content-probing validation, then extended to a second modality for research completeness.
- **[REVISED]** A minimal command-line interface that prints the evidence-backed recommendation card is moved into core MVP scope (previously an optional extension). With five to six engineers available, its cost is small relative to its value for demonstration, grading, and the human evaluation study in Section 18.4, which needs a real artifact for participants to react to.
- The final system prefers reproducibility and measurable gains over adding many loosely integrated features.

20.2 **[NEW]** Milestone M0: Compute-Budget Validation Spike

Before any subsystem is built at scale, the team runs one full evidence-collection-plus-probe cycle on approximately 10 real datasets, on the target hardware (MacBook Air M1, 8 GB RAM), and records:

- wall-clock time per candidate for retrieval, evidence resolution, sampling, and probing;
- peak memory usage per stage;
- failure rate of sample acquisition per repository.

These numbers directly determine whether the 50–100 candidate-pool default (Section 5.1) and the 30–100 candidate range in the retrieval procedure are realistic, *before* the team commits a milestone schedule around them. If per-candidate probe cost exceeds a set budget (e.g., 15 seconds), the candidate pool size or probe depth is revised at this point rather than discovered during M4–M5. M0 is scheduled for the first 1–2 days of the project, in parallel with environment setup (M1).

20.3 **[NEW]** Team Ownership (Five to Six Engineers)

21 Risks, Limitations, and Safeguards

- **Licensing risk.** Automated license parsing can be wrong or incomplete; mitigated by treating license fields as claims subject to Evidence Resolution (Section 8.2), never as legal clearance.
- **Privacy risk.** Sample inspection could surface sensitive content in poorly documented datasets; mitigated by restricting to publicly listed datasets and sanitizing any logged samples.
- **Popularity feedback loop.** Left unchecked, ranking would reinforce already-popular datasets; mitigated by the long-tail pool and measured directly via EDI (Section 12.1).

Subsystem	Scope	Consumes / Produces
Retrieval & Ingestion	Source adapters, canonical schema, BM25/dense indices, M0 spike	Produces: candidate pool
Evidence & Probing	Evidence Resolver (Alg. 1), Cleanlab integration, fingerprinting, cold-start transfer estimate	Consumes: candidate pool; Produces: resolved evidence + utility
Ranking & Diversification	Multi-objective model $R(D \mid Q)$, GIST-inspired reranking, long-tail pool, EDI audit	Consumes: evidence + utility; Produces: final set
Evaluation & Calibration	Hidden-gem/robustness benchmarks, ablations, ECE/Brier, human study (owns Section 18.4)	Consumes: full pipeline outputs
Reporting & CLI/UI	Explanation card rendering, minimal CLI, documentation, final report	Consumes: final set + evidence

Table 2: Suggested subsystem ownership for a five-to-six-person team. Interfaces between rows are explicit data contracts (e.g., the schema of a “resolved evidence” record) agreed before implementation begins, so subsystems can be built and tested independently.

- **[NEW] Compliance risk in automated sampling.** Programmatic access to third-party platforms can violate API terms if unmanaged; mitigated by the per-source compliance checklist (Section 16.3) and caching-first fetch policy.
- **Compute risk.** Probe-based utility estimation is the most compute-intensive stage on constrained hardware; mitigated by the M0 validation spike (Section 20.2) and the transfer fallback (Section 11.1), which avoids forcing every candidate through a full probe.
- **Overclaiming risk.** A system that reports high-sounding confidence without empirical support is worse than one that reports honest uncertainty; mitigated by the calibration requirement (Section 12.1.2), which gates any “High” confidence label on measured ECE/Brier performance.

22 Expected Project Contribution

The project’s contribution is not a new language model or a generic RAG interface. It is an integrated methodology for moving from dataset retrieval toward evidence-driven dataset selection, with a specific, independently citable technical core: a **Dempster–Shafer evidence-resolution algorithm for multi-source dataset trust claims** (Section 8.2), evaluated for both bias reduction (Section 12.1) and calibration, and validated against a human trust study (Section 18.4) rather than automated metrics alone. This combination — a formal fusion rule, a measured bias-reduction claim, and a human-validated trust claim — is what distinguishes the project from a re-application of DataFinder, Cleanlab, Croissant, and GIST, and is the part most suitable for framing as a short paper or workshop submission beyond the capstone deliverable itself.

23 Expected User-Facing Result

For a query such as “I want to build a CNN model for classifying plant diseases from leaf images; I need a multi-class dataset,” the system returns a recommendation set with different roles (best overall, highest quality, most efficient, hidden gem, notable alternative). Every recommendation includes source links, calibrated evidence strength, main reasons, limitations, and the specific factors that caused it to outrank or differ from nearby alternatives.

24 References and Research Sources

References

- [1] Lin, V. et al. “DataFinder: Scientific Dataset Recommendation from Natural Language.” *WSDM 2023*. <https://arxiv.org/abs/2305.16636>. Official implementation: <https://github.com/viswavi/datafinder>.
- [2] Mazumder, M. et al. “DataPerf: Benchmarks for Data-Centric AI Development.” *arXiv:2207.10062*. Official repository: <https://github.com/mlcommons/dataperf>.
- [3] Northcutt, C. et al. “Confident Learning: Estimating Uncertainty in Dataset Labels.” *NeurIPS 2021*. Implementation and documentation: <https://github.com/cleanlab/cleanlab>, <https://docs.cleanlab.ai/>.
- [4] MLCommons. Croissant Dataset Metadata Format Specification, including the Croissant 1.1 provenance and usage-policy extensions. <https://docs.mlcommons.org/croissant/>.
- [5] Zadimoghaddam, M. et al. “GIST: Greedy Independent Set Thresholding for Max-Min Diversification with Submodular Utility.” *arXiv:2405.18754*, NeurIPS 2025. <https://arxiv.org/abs/2405.18754>.
- [6] Google Research. “Introducing GIST: The Next Stage in Smart Sampling.” <https://research.google/blog/introducing-gist-the-next-stage-in-smart-sampling/>.
- [7] Public reporting and the Data Provenance Initiative on dataset licensing and provenance quality in widely used public ML datasets, used to justify verification and uncertainty rather than treating repository license fields as ground truth.
- [8] Google Dataset Search research and dataset discovery literature, used as contextual background for large-scale dataset discovery and heterogeneous source indexing.
- [9] Hugging Face Hub dataset documentation, Dataset Cards, and dataset viewer documentation, used as practical evidence-source patterns for dataset context and sample inspection.
- [10] Shafer, G. *A Mathematical Theory of Evidence*. Princeton University Press, 1976.
- [11] Dempster, A. P. “A Generalization of Bayesian Inference.” *Journal of the Royal Statistical Society, Series B*, 1968.
- [12] Guo, C. et al. “On Calibration of Modern Neural Networks.” *ICML 2017*. Used as the methodological basis for the reliability-diagram and Expected Calibration Error evaluation in Section 12.1.2.

Appendices

A Practical Repository Structure

```
capstone-dataset-intelligence/
|-- data/
| |-- raw/ # source snapshots where permitted
| |-- normalized/ # canonical dataset records
| |-- evidence/ # evidence ledger records (incl. Bel/Pl/K_total per claim)
| |-- fingerprints/ # cached sample-level features
| '-- benchmarks/ # queries + relevance judgments (hidden-gem, robustness)
|-- indexes/
| |-- bm25/
| '-- dense/
|-- src/
| |-- ingestion/ # per-source adapters + ToS/compliance metadata (Sec. 17.2)
| |-- normalization/
| |-- retrieval/
| |-- evidence/ # Dempster-Shafer resolver (Algorithm 1)
| |-- probing/ # incl. cold-start transfer estimator (Sec. 12.1)
| |-- ranking/
| |-- diversification/
| '-- reporting/ # explanation cards + CLI (moved into MVP scope)
|-- experiments/
| |-- baselines/ # incl. LLM zero-shot baseline (Sec. 20)
| |-- ablations/
| |-- robustness/
| |-- calibration/ # NEW: ECE / Brier / reliability diagrams
| |-- bias_audit/ # NEW: EDI / Gini
| '-- long_tail/
|-- human_study/ # NEW: protocol, consent notes, Likert data (Sec. 19.4)
|-- configs/
|-- scripts/
|-- tests/ # incl. resolver unit tests (robustness corruption cases)
|-- requirements.txt
'-- README.md
```

B Minimum Viable System vs. Research System

Minimum (M1–M6)	Viable System	Full Research System (M7–M8+)
Single modality, retrieval, metadata evidence, basic Dempster-Shafer resolution with default priors, greedy diversification, CLI output.	hybrid re-	Second modality, tuned reliability priors via calibration, full ablation suite including LLM baseline, EDI/Gini bias audit, human trust study, GIST-inspired reranking, tuned $R(D Q)$ coefficients.

C Implementation Stop Conditions

- If dense retrieval does not beat the lexical baseline, stop tuning downstream ranking and revisit corpus normalization/query formulation.

- If sample acquisition is unreliable across repositories, keep metadata/provenance evidence as the fallback and clearly label content evidence as unavailable (or transferred, Section 11.1).
- If probe models consume excessive local compute, reduce the candidate pool and sample size before reducing evaluation rigor — guided by the M0 measurements (Section 20.2).
- If long-tail diversification decreases human relevance materially, tune the exploration quota rather than forcing low-quality novelty.
- If license/provenance evidence is contradictory, the system lowers confidence or flags the dataset (the CONFLICTING state, Section 8.2) rather than inventing a resolution.
- If the full multi-modal pipeline becomes too large for Capstone 1, freeze one modality for the end-to-end experiment and retain the second modality as a validated extension.
- **[NEW]** If the reliability priors in Table 1 produce poor calibration (high ECE) even after tuning, fall back to reporting raw evidence counts and source lists without a synthesized confidence label, rather than presenting an uncalibrated number as if it were meaningful.

D **[NEW]** Worked Example: Evidence Resolution for a License Claim

Consider the claim $c = \text{“license of dataset } d \text{ is CC BY 4.0,”}$ with three sources:

- s_1 : bare repository metadata field, asserts true, $r_1 = 0.60$.
- s_2 : official dataset card, asserts true, $r_2 = 0.70$.
- s_3 : an unverified community mirror README, asserts *false* (claims “All Rights Reserved”), $r_3 = 0.35$.

Step 1 — BPAs. $m_1(\top) = 0.60$, $m_1(\Theta) = 0.40$. $m_2(\top) = 0.70$, $m_2(\Theta) = 0.30$. $m_3(\perp) = 0.35$, $m_3(\Theta) = 0.65$.

Step 2 — Combine m_1, m_2 . No conflict yet since both support $\top$: $m_{12}(\top) = m_1(\top)m_2(\top) + m_1(\top)m_2(\Theta) + m_1(\Theta)m_2(\top) = 0.60(0.70) + 0.60(0.30) + 0.40(0.70) = 0.42 + 0.18 + 0.28 = 0.88$. $m_{12}(\Theta) = m_1(\Theta)m_2(\Theta) = 0.40(0.30) = 0.12$. $K_{12} = 0$.

Step 3 — Combine with m_3 . Conflict mass: $K = m_{12}(\top)m_3(\perp) = 0.88 \times 0.35 = 0.308$.

$$m_{123}(\top) = \frac{m_{12}(\top)m_3(\Theta)}{1 - K} = \frac{0.88 \times 0.65}{1 - 0.308} = \frac{0.572}{0.692} \approx 0.827$$

$$m_{123}(\perp) = \frac{m_{12}(\Theta)m_3(\perp)}{1 - K} = \frac{0.12 \times 0.35}{0.692} = \frac{0.042}{0.692} \approx 0.061$$

$$m_{123}(\Theta) = \frac{m_{12}(\Theta)m_3(\Theta)}{1 - K} = \frac{0.12 \times 0.65}{0.692} \approx 0.113$$

Step 4 — Interpret. $\text{Bel}(\top) \approx 0.827$, $\text{Pl}(\top) \approx 0.940$, $K_{\text{total}} \approx 0.308$. Since $K_{\text{total}} = 0.308 > \kappa_{\text{high}} = 0.30$, Algorithm 1 returns CONFLICTING *despite* $\text{Bel}(\top)$ being high — correctly surfacing that one credible-enough source actively disputes the claim, rather than averaging the dispute away. This is the behavior a simple weighted mean cannot reproduce: a naive average of $\{0.60, 0.70, (1 - 0.35) = 0.65\}$ would report ≈ 0.65 and hide that the sources are in outright disagreement rather than moderately uncertain.

#	Gap identified in review	Resolution in Version 2
1	Evidence states were assigned heuristically, no combination rule.	Dempster-Shafer resolver, Algorithm 1, Section 8.2.
2	Bias/diversity claims were asserted, not measured.	Exposure Disparity Index and calibration metrics, Section 12.1.
3	No comparison against the “just ask an LLM” alternative.	LLM zero-shot baseline added as ablation #1, Section 19.
4	No defined behavior when a dataset cannot be sampled at all.	Transfer-estimate fallback, Section 11.1.
5	Trust/explanation quality was a design principle, not evaluated.	Human trust study, Section 18.4.
6	M1-Air compute feasibility of the 30–100 candidate pool was assumed.	Milestone M0 validation spike, Section 20.2.
7	No task ownership across the 5–6 person team.	Subsystem ownership table, Section 20.3.
8	Automated sampling had no explicit ToS/compliance step.	Legal/ToS compliance checklist, Section 16.3.
9	CLI/demo interface was an optional extension (M6).	Moved into core MVP scope, Section 21.1.

E [NEW] Changelog: Version 1 to Version 2

Final project definition. This system is a local, research-oriented dataset intelligence pipeline that retrieves candidate datasets, resolves their evidence through a formal, testable fusion rule, estimates task-specific usefulness (with an explicit fallback when direct measurement is infeasible), measurably mitigates popularity bias, diversifies recommendations, calibrates its own confidence, and validates its explanations against human judgment. Retrieval remains necessary, but retrieval is no longer the endpoint — and the system’s claims about evidence, bias, and trust are now each individually falsifiable.